Stars Ask Neil To Save The Universe

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis This Star Talk "Celebrities Ask Neil" Special Edition episode serves as a bridge between rigorous scientific inquiry and popular culture entertainment. Hosted by physicist/celebrity Neil deGrasse Tyson alongside Gary O'Reilly, the segment utilizes questions from Hollywood figures (including Steven Spielberg, Bill Prady, Hassan Minhaj, Josh Scherer, Ricky Lindholm, Emily Blunt/Britt Day?, Kevin Sussman) to demystify complex concepts. The core thesis is that scientific principles—ranging from quantum mechanics and atomic structure to cosmology, genetics, and space physics—are not confined to the laboratory but can be applied directly to debunk cinematic misconceptions, explain everyday phenomena like salt chemistry, and address critical global challenges such as climate change resilience in agriculture through genetic modification.

2. Key Arguments & Data

3. **Atomic Structure & Quantum Mechanics:** Atoms consist mostly of empty space; electrons do not collapse into nuclei because they exist as wave-particle duality entities confined by their wavelength (Heisenberg Uncertainty Principle). The Rutherford scattering experiment revealed this vast emptiness ("crackerjack kernels"). While quantum tunneling allows particles to cross barriers instantaneously, the "gap" between nucleus and electron is not a vehicle for human faster-than-light travel. Spielberg’s suggestion of using frequency adjustments on electrons hints at energy levels (magnetic field excitation) rather than simple orbitals; he conflates orbital structure with entangled particle manifestation without fully grasping causality limits in communication or transport.
4. **The Multiverse:** Bill Prady (*Stuart Fails to Save the Universe*) argues that merging quantum physics and general relativity mathematically suggests multiple universes exist ("the shotgun marriage" of large/small scale). While theoretical evidence points toward existence, interaction is currently unproven and potentially hazardous due to potential variations in physical laws across fluctuation bubbles.

5. **Salt Chemistry & Nutrition:** Pink Himalayan salt derives its color from iron oxide impurities (rust), not the geological composition of the mountains themselves; it contains no nutritional advantage over refined white table sodium chloride ("overly processed" argument). While useful for flavoring, consuming it based on mineral claims is an overstatement compared to dietary vegetables/fats. The etymology notes Roman soldiers were historically paid in salt (*salary* = payment by weight/salt), though modern usage of "salt money" does not change nutritional reality.
6. **Plant Genetics & GMOs:** Nature frequently performs genetic modification via crossbreeding (e.g., apple trees grafting different varieties on one stem; corn size variation). Almost all food is genetically modified naturally or artificially ("almost nothing we eat isn't"). Proposals include "Chocolate Broccoli" to entice child nutrition and engineering crops for climate resilience (drought resistance, frost tolerance) rather than just yield. A controversial example cited involves inserting sub-zero temperature genes from cold-water fish into tomatoes ("Frankenmado"), highlighting biological interconnectedness despite public objection.
7. **Space Physics Misconceptions:** *Common Sci-Fi Errors:* Sound does not travel through vacuum (lightning seen but thunder heard later on Earth; explosions in movies are often silent or audible). Weightlessness is only felt during freefall/zero-G when engines are off, whereas firing rockets creates weight/G-force (*Ad Astra* depicted incorrect physics compared to *The Expanse*).
    - **Time Travel & Paradoxes:** While forward time travel (dilation) occurs via relativity/speed of light constraints (Richard Gott's work), backward time travel risks paradox problems preventing causality. Stephen Hawking's "Chronology Protection Conjecture" suggests nature prevents killing ancestors to avoid self-destruction loops (*Big Bang Theory* reference).
    - **Alien Encounters:** Tyson denies personal contact but acknowledges the possibility of unacknowledged encounters (Fermi Paradox logic: they might be watching without revealing themselves due to advanced capability or humor rather than greed. Potential universal communication via math/Pythagoras theorem and emotional resonance, bypassing language).

8. **Agriculture & Climate:** Crop yields are suffering from heat/drought; genetic engineering is necessary for early maturing seasons (double yield) or cold/salt water survival to ensure food security during climate crisis escalation.

9. Actionable Takeaways

10. **For Science Advocacy and Pop Culture Integration:** Leverage the comfort zone celebrities create around "geek" interests in media (*Big Bang Theory*, *Family Guy*) to introduce scientific accuracy (e.g., weightlessness physics) alongside entertainment, as seen with *Stuart Fails to Save the Universe*.
11. **Consumer Awareness on Health Claims:** When purchasing household ingredients like salt, recognize that color variations imply impurities rather than superior nutrition; do not rely on marketing-driven health claims for processed mineral content. Prioritize whole foods (vegetables) over "specialty" salts ("lazy salt").
12. **Educational Engagement with Movies:** Use film analysis as a learning tool to correct common misconceptions about sound in space and spacecraft dynamics, citing correctly portrayed science shows like *The Expanse*. Encourage viewers to question how physics is depicted versus reality.
13. **Agricultural & Climate Preparedness:** Support or advocate for genetic modifications that help plants adapt to climate stress (heat/drought resistance) rather than viewing GMOs as "unnatural," given that nature modifies organisms daily ("nature's own laboratory"). Focus on child-friendly vegetable flavors and nutrition through flavor enhancement strategies.
14. **Maintain Perspective on Theoretical Physics:** When discussing speculative concepts like quantum tunneling or FTL communication, differentiate between theoretical particle behavior (wave function collapse) and actual macroscopic travel capabilities which remain unachievable due to constraints of causality and energy requirements.

Full Transcript

I love Celebrity's Ask Neil. Yeah. Okay. It's just a reminder that, you know, they're people too, and they have a geek-soft thing to do. They have cosmic curiosity, just like the rest of us. And they got really personal, those people. I gotta tell you, I was offended by some of those questions. Stephen Spielberg's question, that offended you? Yes. What a nut he is. Coming right up, Celebrity's Ask Neil, Star Talk Special Edition. Welcome. to Star Talk, your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk, special edition, which means I got Gary O'Reilly. Hello. Hey, Gary. Okay, so today we have one of our occasional installments of Celebrities Ask Neil. Oh. Those are fun. Yeah. And generally, you know, if I find myself around town, on a movie set. As you do. As can happen on occasion. I'll just pull out my smartphone and see if they have a question for me. If one of the celebs have a question for me. Because everybody's got a question about the universe. Right. And so we gather enough of them on the shelf and bring them out as an entire episode. You know, I do the same thing when I'm on set and the answer always comes back. How did you get past security?

So it's kind of like cosmic queries for celebs. Oh, that's cool. That's a way to think about it. All right. Do we charge them like Patreon? No. Give them a five dollars. Five dollars. Top tier. You forget the five dollars. These are stars we're talking about. All right. So first up, let's see who we've got on the docket. Okay. Neil, Steven Spielberg here to ask you a question. I'm not a physicist and I don't have any real knowledge of quantum mechanics or any of this, but I am curious about what is the gap between, you know, an electron. What is that gap where the nucleus, you know, is separated and you've got this gap? And is that gap a way to travel, to be able to defy having to go the speed of light to get somewhere. Wow, where did you get to talk to Steven Spielberg? I mean, that was kind of wild. Hey Neil, Steven Spielberg here. Where's that happening? Well, so I had access.

Yeah, we just saw that. So two things. As a StarTalk entity, we were invited to the early press encounters with the actors and producers and writers for Disclosure Day. Nice. And not only that, I also got a red carpet invite. Oh,

cool. Yeah, that was fun. That's yours is in the mail. Yeah, well, no, I got the blue carpet in bite. I got the cease and desist. Exclusions alone. Yeah. So here we caught up with him when we were chatting with him about the film, putting together a whole episode on what the film was about and aliens. Cool, cool, cool. Yeah. So he asks a perceptive question that most people don't even think about. OK. All right. So...

In an atom, there's empty space between the nucleus and the electron. What's the charge on the nucleus? It's a positive charge. That's where all the protons are. What's the charge on an electron? It's negative. It's the opposite charge. They attract. Of course. Why doesn't the electron slam into the nucleus? Well, they want to, but it's kind of a Montecue and Capulet type thing. They're being kept apart from one another, even though they love each other dearly. Oh, thank you, Shakespeare, for that. I'm here in Juliet. So the distance you took, is that the Max Planck sort of scenario? Oh, the Planck is much smaller. Much smaller, though. But if you want to get a sense of how big that gap is, we have a few kernels of... What's that popcorn they eat in stadiums? A popcorn? A crackerjack? A crackerjack. We have a few crackerjack kernels in the middle of the field.

The volume of the stadium is where the election was. That's amazing. That's that much space. That's incredible. And the first guy who discovered there was that much space, his name is Ernst Rutherford. He loved baseball. He's a New Zealander. New Zealander. Ernst Rutherford. Kiwi. A kiwi. One of your colonies. Thank you for reminding me of our colonial past. Just saying. But I think they still have your Union Jack on their flag. Yeah, that's... You see, when they pee on all their flags and it's still there. Okay. I don't think we actually did that. But... Okay. Don't let it get away with a good story. Alright, try it. So, he was the first to discover all this empty space. And it's rumored, just rumored that the next morning, who's afraid to step out of his bed out of fear that he would fall through the floor. Nice. That was one hell of a dreamy hell of a house. I was gonna say the night before, he got high, I don't know what. Realizing that most of matter was just simply empty space. Alright, so that's one factor. That alone has nothing to do with faster than light travel, but it is an intriguing fact that the electron does not collapse into the nucleus. So what Spielberg is doing there, is he foreseeing?

a movie that he wants to make to explore access through the gap. I want to call it space, but not in the classic tone. It wouldn't be the first time that happened with me. Yeah. Okay. I got a phone call from Seth McFarland and said, Neil, I would be in town. Can we do lunch? And I definitely made lunch for Seth. And we had lunch and he asked me all these questions about the Big Bang, the age of the universe and everything. And then six months later, there's an entire Family Guy episode. where Stewie's time machine takes him back to the big bang. To the big bang, yeah. And at the end it says, science consultant Neil Gretzky. Nice. It should have said lunch conversation with Neil Gretzky. So, there's that gap. So here's why the electron can't just fall in. All right. Because the space occupied by the electron is set by its wavelength because it's also a wave. The wave particle duality. If the wave can't fit, In a smaller space, it's not going there. It's going to occupy the bigger volume. Cool. Okay. So, just something to keep in mind. You can't just stick it anywhere. Right. Okay. So now, what he might have also been hinting to was quantum tunneling. Right. Right. Okay? And quantum tunneling, when you have entangled particles, they're entangled and they know about each other and when you observe one of them, the other one manifests at faster than the speed of light. So I think he was... He was kind of conflating the two there. A little bit, but I think his heart's in the right place. Because he's thinking this. What happens if you change the frequency on the wavelength? Of the what? Of the electron. That's what he's saying. You can adjust the energy levels of the electron with a magnetic field. You can excite them or de-excite them. You can do things, but the structure of the nucleus is really what's responsible for all of what the electrons will do outside of it. So, and so badly did we want atoms to resemble solar systems. So badly, because it's easy to think that... To think you have a center like the sun and then other things. It's just that all the way down. It's just all the way up and all the way down. It's Mother Nature replicating. So badly did we want that, that we call the electron positions orbit Tolls yeah, instead of orbits right orbitals, so they're statistically configured Places where you would find the electron around the nucleus the point is when anything tunnels in quantum physics It goes across a barrier instantaneously all right, and when you have two quantum entangled particles they manifest instantaneously Okay, as far as any of our measurements and calculations go positions. Okay, one is here and was here. Yeah, right so How close are we to harnessing that? We got top people

working on doing, trying to do just that. How long? They've been working for like at least 10 years. It's slacking. Quantum physics is hard stuff. Step it up, quantum physicists. Step it up. So the electron has a field. Yeah, so does the proton. So is it the fields that are interfering that keeps them apart?

No, no, it's the fact that you cannot physically put an electron in that space. It's no different from why a microwave can't get out of a microwave oven. Right. Because the holes in the door are smaller than the wave itself. So the wave came, doesn't even know the holes there. Right. So anyhow, I think you're right. He might be thinking of a movie or something. Well, you wouldn't be a surprise. That'd be kind of cool. It'd be kind of cool. Yeah. Nothing wrong with that. See how he wants to take that forward. Right, do we all do we three get a an assist if the film gets made? Oh get a mention honorable mention Really and the first time I ever met him it was in this office Oh really came by the IP services apparently he made a request brought one of his sons in Who was just at that age maybe 14 13 nice? Yeah, yeah, and that's it and we chatted and that's when I learned Because I asked go ahead that ET was a vegetable not an animal. That makes sense ET was plant-based. I believe that that's why ET had that magic finger that can make plants come back to life Yeah, that finger he could touch a plant and the plant rejuvenate So there it is. That's very cool. Okay. Hi, I'm Ernie Carducci from Columbus, Ohio I'm here with my son, Ernie, because we listen to StarTalk every night and support StarTalk on Patreon. This is StarTalk with Neil deGrasse Tyson. All right, next up. Hi, I'm Bill Prady, co-creator of the new HBO Max series, Stuart Fails to Save the Universe, and I have a really critical question for you, Neil. And this is, everything depends on it. Are multiverses real?

Or should that be, is the multiverse real? Or should it be, are there multiple universes? One of those three questions is critical. Okay, so you know what Bill Brady is? Other than, so he's got this new show. Stuart fails to save the universe. The premise of that is this character, Stuart, is he breaks the space-time continuum and is accidentally wandering into multiple universes. Yes. Okay? Now, what did he break? A device made by Sheldon from the Big Bang Theory. The Big Bang Theory. So Bill Prady co-created the Big Bang Theory with Chuck Lorre. Chuck Lorre. Who also created Two and a Half Men. And every other thing that you've seen on TV. Every

other thing. Yeah, pretty much Chuck Lorre. And so I'm known to them, perhaps through Star Talk, but even in the day, I was invited to give two cameos on the Big Bang Theory. Oh, very nice. One in Season 3 and one in Season 7 or 8, the last season. Nice season. Yeah. That's cool. Yeah. That's very cool. Well, first cameo, I argue with Sheldon about Pluto's planet status. Okay. Okay. Then, the second one, I am settling scores by phone, okay? I'm calling people up in sequence.

and one of the lines that was written on the spot was not in the script. On the spot is, all right, who else needs a degrass whoopin'? Oh, that's funny. That was good. That's a good one. That's a good line. All right. So, anyway, that's who he is. And so apparently Stewart, who's a character from the Big Bang Theory, perhaps the most awkward geekiest character among the geeky characters is Stewart. He owns the comic book store. Yes. The comic book store. He needs to know if there's... a real multiverse. Otherwise, the premise of the whole show fails. Pretty much. Okay. When you do the math, merging quantum physics and general relativity, the physics of the small and the physics of the large, there was a point in the past when the large was small. That was the big bang. There's the shotgun marriage. When you do that at a math level beyond my...

Reach, that's what we need Brian Green for. Multiple universes pop out of the equations. It wasn't like we try to pull it out. You didn't try to make it happen. You got it for free. You got it for free. And you have quantum fluctuations. Every time you have quantum fluctuations, things happen. And if you have quantum fluctuations, when the whole universe is the size of a fluctuation, the whole universe becomes one of those fluctuations. Exactly. And there are variations on the multiverse. In fact, I invite the listener reader to see the start talk book Cosmic Queries. There's a whole section on there on how the universe will end. And it talks about multiverses. There's a whole discussion in there on that. So, Bill Prady. Bill Prady, all theoretical evidence points to the existence of a multiverse. Whether we will ever one day be able to interact with it by some backdoor method remains to be seen.

But if it's a fluctuation, other things would have fluctuated as well, such as the laws of physics. Very dangerous. Which means we don't want to go

there. Correct. Yeah. Very dangerous. So, Bill, take it and run with it. May your show... Send the check here. Send the start talk here. Check nice. So, there it is. So, Bill, keep up the good, fun work. You finally gave the geekiverse a Hollywood voice. Yeah, man. I don't know how else to say that. No, that's a good way to say it. I think he's part of the reason from, I'll say, a pop culture and general culture framing that people are far more open to scientific information because of his show. And as long as it's been on the air. That shows created comfort zone for geeks to exist in. Yes. That's the better way to say it than the way I said it. It's a comfort zone. It's a comfort zone for geeks. And while still celebrating their brilliance, you know, you can still make fun, but no one at the end of the day, you don't get the sense that anyone's feelings are getting hurt or they're getting exploited for their lack of social skills.

It's kind of a celebration of this who I am, deal with it. And I'll make a mistake here, but I'm self-aware and I'm shy when others are not, but that's just what I am. He's been crafted. And he doesn't talk down to make the jokes. Good point. He actually makes, he leaves the science to science and then writes a sitcom around that. Ooh, good thinking about that. Which is really cool. All right, next up. Hey Neil, it's comedian Hasan Minhaj. Look, I got a question for you. Why is pink Himalayan salt pink? Like are the Himalayas pink or is the rock pink? Or am I eating a piece of the Himalayas? What's going on? So first of all, I was duly informed by who I don't remember that if you have more than one kind of salt in your cabinet, you're rich. That is not true.

I can vouch for that for certain, okay? Because I have coarse sea salt, I have pink Himalayan salt, I have black salt, okay? And then of course, Mortons. Evidence of how much we're paying you. No, no! That's what I was saying! Nope, no. I'm just saying. What the people don't know is they pay me in salt. Like it's 200 BC. Back in the day. Back in the day. That's where the word salary comes from, salt. Oh, cool. That's very cool. When the Romans paid their soldiers in salt. In salt? Yeah. Nice. So I'm just saying, if you go up in the hood, nobody's got more than one kind of salt in the cabinet. Now, that's true. I'm just saying. That's true. That's because, you know, black people know how to flavor their food. They don't need five kinds of salt. They got other things. We got stuff like fat back. All right? You don't need

no damn Himalayan salt when you got fat back. Okay? But do you know what fat back is? No. That's what I was wondering. No, no, just thinking, I don't want to ask. I don't want to ask. I'll tell you.

A fat bag is a slab of a pig that doesn't have enough meat to even slice it into bacon. Which is itself half fat, right? This has less meat than a fatty bacon does. And you pop that in your... But used correctly, it will flavor the hell out of anything. Especially collard greens. You just throw it in there. Yeah, and the vegetarians be damned. Yep. You don't let them know that it's in there. They just go, oh my God, I've never had greens this good. Oh my God, what's your secret? I'm just, oh, it's just wonderful. I've just never had greens so flavorful. Oh. So I think fatback is the between state of existence of bacon and lard. Yeah, basically. What do you think of that? Yeah. That's a pretty accurate question.

Yeah. All right. I'm not to be confused with the pork belly. Yes. Which is just the opposite of the fat back. Yeah. Okay. Yeah. But go ahead. So there's salt in so many places on earth because you just need a lake that has been around long enough to become salty and then the lake evaporates and leaves a salt residue and then it gets buried. And we have on another episode we learn that the crystalline state of atoms or molecules is the lowest energy state that it can Acquire so that once it hits that state is that way basically forever basically All right, so that's why you can dig up salt. That's why salt is in mines where the salt's been there for a million years. Yeah, there it is. Why is it pink then? Says the man who today is wearing a pink my favorite shirt look good in it. I like it Yeah, I don't agree. You don't like it. No, what's wrong? You don't have to wear it So it comes from impurities in the pure crystal lattice of the salt molecule, which is sodium chloride. And in this case, the impurity is an iron oxide, so infusion. Oh, okay. Let me call that. Okay.

It's rust. It's rust. You're eating. You're eating rusty stuff. Oh, rusty ass stuff. That's what you're eating. So you're telling me, would you pay for that stuff? Yes. How would you pay for that rusty ass stuff? That's making me feel slightly better. It's centuries of crushed flamingos. Oh, I like flamingos. I like that. Yeah, but they have to get pink. First, they gotta eat the shrimp and the krill. They gotta eat that first, then they become pink. Then they get

crushed. Thank you for that new hypothesis. First they have to hike to the Himalayas. Fomingos ain't going to the Himalayas, I promise you. Oh, hilarious. These are trace amounts of this iron oxide that's in there. And you can ask, well, how much is in there? Right.

because it's a good source of iron. Yes, it is a source of iron, but so many other foods you eat in a day, especially fresh vegetables, are sources of iron, far beyond what you're getting in the salt. So, while it might look pretty in a transparent salt shaker, salt grinder, which I have at home. Of course. Of course. It wouldn't be so gross to put salt in it. To buy it on the premise that you are somehow healthier for it is an overstatement of its nutritional value. You're ripping yourself off. To you. So salt is overly processed and pink Himalayan salt is just lazy salt. Correct. So other salts might have some tinge to them, but they refine it.

to remove the other minerals. Exactly. And you left with pure salt, sodium chloride. Yeah, because everybody wants, you know, for table salt, it's supposedly, you know, white is better, you know, because like, all right, let's be real, right? Anyway. But if you were, if you went back in a day and you got salt, it was gray. Yeah, I have gray salt as well. Yeah, your salt was gray. And that's like the most natural salt you're gonna get. So, yeah, and I saw Menage, we love him. And I was on his podcast. And I just thought I can't exit his podcast. He was on your live show. Yes, he was. He was on Star Talk Live. Really smart guy. And I first saw him through his appearances on The Daily Show. Oh, cool. Yeah. Yeah. And Hassan, something else about your pink salt from Himalayas. Other salt that is white or otherwise colorless, if you look at it.

colonized that Himalayan salt and went in there and took that Himalayan salt and said, listen, y'all pink salt, y'all gonna understand something. Chuck is still in therapy. Nobody seems quite happy. He's been completely happy. Anytime I can get a chance, that's all I'm saying. Next up. Hey, my name's Josh Scherer and I'm the host of Last Meals on Mythical Kitchen and I have a question for Neil. Since the beginning of time, humans have been manipulating plants via crossbreeding and later GMOs. What of the plants that we eat do you think is most currently lacking and needs an update? Oh, I love that. I love that. There's hardly any food in the grocery store that has

not been genetically modified from its original form. There's almost no food on Earth that has not been genetically modified. That's basically true. There are a few things like wild caught tuna that's not been genetically modified. But almost everything else we have modified it. And even if we haven't modified it.

Nature genetically modifies itself. Just think of corn. Just think of your original corn. Ritual corn's this big. Now they're this big. People say, oh, I want natural corn. I don't want GMO corn. Oh, you want natural corn? You want some malnutrition? Some corn that you could eat, 18 of them. And you still be hungry. Exactly. So the only difference is GMOs, what we think of as GMOs, are accomplishing the same thing that we did. in artificial selection, but we just do it in a laboratory. And both of those are what nature does all by itself. How else are you going to get from single cell organisms to elephants? Absolutely. Somebody's doing some serious genetic modification. But sometimes you can just leave plants and they will do it themselves. They'll actually crossbreed. Flowers will do it. Do it. Do it. Do it to satisfy. That's it. We'll do it.

There's a 70s disco. Oh man, that's hilarious. Yeah, they'll cross me with whatever they might. In a limiting case, I'm told by experts in the field that there are apple trees where different trunks of the tree are differently grafted so that the same tree can give you multiple kinds of apples. Oh, that makes sense. Yeah, yeah. That's cool. That's yeah, that's like. I love that. I'm gonna make my own cornucopia. Right, exactly. I want a Fuji Gala Granny Smith tree. So I have to pick a plant that I think is in dire need of some more genetic modification, right? Okay, let me think about that. Do you have it? Are you thinking one? Yeah, I'm trying. I'm just racking my brains. Mainly it's vegetables to give them a flavor that allow Kids do want to eat them. Oh Oh, that's good. Chocolate flavored broccoli. Chocolate broccoli? We'll call it chocolate. We're in. There you go. It's chocolate broccoli. It's chocolate broccoli. Chocolate broccoli. It's a reference from 1980s Saturday Night Live. All right. True. Very true. Very true. I wasn't aware of that. No, that's right.

So, let's agree on that, that broccoli, the famous food to be hated by children, we need to get them to like that food, and we need to infuse it with

something we know they will like. But they will like it. So, let's get the chocolate gene. Right. Either that or we work on the jelly belly, multiple flavors, and then you can choose which one you can. In the broccoli? Yeah. Okay. Oh, oh, it just says, different florets. Right. How you like my vocabulary? I'm good with that. Different florets would have different flavors. So you'd have multi-colored broccoli with different flavors. Now you got it. The kids would be eating that up. Okay, there it is. We can't have kids eating rainbow broccoli. Yeah, on a serious note, a whole other reason for GMO plants, which is more broadly called genetic engineering, is you can genetically engineer plants, for example, to have two crops in one season. If it's early maturing. And then that doubles your yield. Or you can have one that can survive colder temperatures. Or drought resistant. Exactly. So they don't evaporate as much water out of their surfaces. So there are many reasons why one would do this. And by the way, that's going to become far more necessary as the planet.

taking native species of plants and trees from heat, very hot areas that are drought resistant. And taking them over here. Growing them and developing them, so yeah. They're way ahead of the story. Yeah, it has to be because crop yields are already suffering because of climate crisis and it's only going to get worse. There was a controversial GMO where they got some gene from a fish that thrives in sub-zero temperatures of salt water, so like 30 degrees, 31 degrees, and they put that in a tomato, which prevents tomatoes from freezing in the first frost. Right, the frost, they're frost resistant. Frost resistant, correct. People freak, does it? I don't want my fish in a tomato, and rather than say, oh my gosh, all life is connected. Exactly.

This, that's... No Frankenmado! Not only are fish our ancestors, there's an earlier ancestor that's the ancestor of both animals and plants. Correct. And green plants. Yes. So, what a celebration of life that actually is. Yeah, yep. All right. Next up. Hi, I'm Ricky Lindholm. I play Ramona on Stuart Fales to Save the Universe. And here is my question. What is the most misunderstood concept in the universe? And why do people keep getting it wrong? Ooh, love that. So, Ricky Lindholm was an intermittently recurring character in the Big Bang Theory. Okay. And now she's a regular in Stuart fails to save the universe. So she plays a most excellent geek in the series. Once again, a celebration of a culture, really. The geek of verse. Yeah.

Yeah, completely. So, that's an excellent question. And I would say, what people persistently get wrong, is that they put sound in space. I'm gonna say, yeah, that's a good one. And they're not giving it up. I go up and slap them and they'll do it again the next time. There's an explosion and you hear the explosion when you see it. Which makes it doubly wrong. You don't even hear lightning when you see thunder on earth. That's correct. And now there's an explosion out in...

Base! And the star blows up and you hear the explosion at the same time? Yeah. With everything that's going on in the world, that's what annoys you? In a movie where they're trying to get their science right. Yeah. Okay. That's right. So Chuck, I remembered you saying that the TIE fighter just happened to have a TIE fighter. Of course. Here in my office, that these make sounds in the vacuum of space. Yes. And it's exactly the same sound they made. in a planetary atmosphere. In a planetary atmosphere, yes. Same with their cannons, you know. And yeah, out in the vacuum of space. So Ricky, that's my biggest issue. Yeah. And I just give it up. People just get it wrong. You mentioned it, just get it wrong. All right, what else? There's got to be something else. Oh, you want another one? OK. You've got more than one inch. OK. Then you want to scratch. OK, here's a more subtle one. OK. People think that any time you are in space, you are weightless.

That is not true. It's not true. If your engines are firing, you have weight inside the craft. And if you want to go anywhere, you're going to fire your engines. So if you turn off your engines, then you're coasting, falling into wherever you're going. And since you're in free fall, only then are you weightless. Right. And the movie, Ad Astra, to getting back and forth to the moon in a matter of hours, rather than three days, three days is the coasting time that it takes to get there, three days. In other words, you fire engines, shut them off, you coast until the moon's gravity... Does the rest. Does the rest, okay? And you're coasting the whole time. You're weightless the whole time. But if you're in a hurry and you don't want to wait three days, you want to get there in three hours, you're burning your rockets the whole time. You're burning your rockets, you are not weightless. And in that movie, they showed the burning rockets, and inside they're all just floating around. It's like, no. No. So what's...

the prohibitor on burning engines all the way to the moon. Is it the weight? How big is your fuel tank? Exactly, yeah. That's the weight. Entirely. So a real future would be filling stations in space. Then you could burn all the way to the filling station and then burn again. Solar powered? Sure, but the farther away from the sun you go, the less intense the sunlight is by the inverse square law of light that we talked about on a prior episode with my butter gun. So, I guess, I guess it's... Google it. Butter gun. Star talk. So, and the show that gets it correct all the time is The Expanse. Yes. They get all their accelerations correct. They do. Two thumbs up. Yep. All right. Next up. Hi, Neil. I've always had this question for you. Have you yourself had an alien encounter? Oh! Oh, we know who that is. She did not identify herself. Yes, but you can tell by her voice. Maybe she didn't need to. Yes, that's how big she is. Mm-hmm.

That she had a little bit of accent there? Oh. No, she doesn't have an accent. You have an accent. Very good. She does not. You have a number here. We can kick your ass. Very good, very good. Watch what you say. This is America Jack. Basha Britt Day. Another Basha Britt Day. So I'm delighted to answer that question. And? I'm not authorized to comment further. There you go. I don't know who this guy is. So do we end up with the Fermi paradox? where the aliens are probably here anyway, or why would they go this far just to... I know. So the question was, have I encountered an alien? The answer is no. But it's not for want of trying. But have you done so not knowing that you had? But that doesn't count because that means that maybe all of us have done so not knowing. Exactly. Then my answer would not be... Yeah, right. If I don't know, then you don't know. No, I agree. I'm just saying, do you think the possibility is that...

We, you, Chuck, everybody watching and listening has had an alien encounter, but has no knowledge of the fact that it happened. Okay, so if that's true for everyone, then it's true for no one. Yeah. And by the way, if aliens are here unannounced and they're observing us, it is because from wherever they came, there are more of them coming. And that is why they don't want to let us know that they are here, which means we're totally screwed. So I don't know these people who are just like, well, they got to be here. They have to, well, you're a dumbass because quite frankly, if they are, that is the end of us because anybody who can reach us is so much

more advanced than we could ever hope to be that when they come here, they're like, yeah, this is ours now.

Only if they behave as we have. Well, my point is this. The only reason they were concealed the fact that they are here is because they do behave the way we do. So we've got grabby aliens. What about if they're not grabby aliens, but they've got a real twisted sense of humor? Now that'd be awesome. How about that? Well then, then I'd be heartbroken that I didn't meet them, to be honest. And I got one for you. I got one for you. Think about us and our sense of humor. I know. And how it can go in different levels. Just imagine an alien like that. I spent a few pages on this in my book. All right. Take me to your leader, a couple of pages. Yeah, I'm about halfway through that. You are, okay. Yeah, yeah. The book is not very long. I'm going to slow you down. The book is like this.

The book is this thick. Anything you want to do to make me feel better? I have a beginner's version of that. So I try to think about what we look like to the aliens. Because they would find it intriguing, I think, that we have these sounds we project forward, vibrating air molecules that reach another person and that they know our thoughts for having done so. If they don't otherwise interact with sound the way we do. That's kind of interesting. Then what might be more interesting is there are some sounds that come out of our mouths that are single syllables that could completely offend another person. They're called curse words. That's weird. A single syllable comes out and the other person is crying because of it. So then we'd have to teach the alien that this is just custom. What's the first thing you do if you're in a...

in another country, you know, their friends. Learn the curse words. Yeah. The first thing that happens is you learn the root words. No, the first thing I do is learn their food, okay? I don't know what countries you go to, what you are after. Okay, I'm taking eating for granted. Okay. But the first words you learn in another language of your day are the swear words. I have done that. Yeah, you gotta learn both so that when you eat the food, you can go, damn! Both. So, all I'm saying is, it would be kinda cool if the aliens also had single-syllable curse words. We'd be visited by cussing aliens. That would just be funny. Gleep, you! If you think about it, if that's got enough technology to transverse... Universal alien speak? Gleep? Gleep. I mean, I

think the technology to transverse large tracts of galaxies, galaxies, universe, and arrive here, they're not going to need to learn a language. They just...

But if they press, if they don't, that's why they have access to be able to interpret everything. Yeah, they did learn the language quickly and they'll just speak English. They land in an English-speaking country. Or they read our minds. You know, that could happen. Because urges and emotions and thoughts don't require words to know that they exist within us. That's interesting. They don't need the... the audible language, they can just go straight through. Right, exactly. And know the feeling and know the attitude. So I've just burned some pages on this. My favorite one is that you befriend the alien. Okay. Then after a certain number of hours, you say, excuse me, I have to lay down horizontally and go semi-comatose for a third of the Earth's rotation. Right. And the alien says, what? And you say, don't you sleep too?

So sleep, this is just what we do. Yeah, it's a weird thing. It's a weird, it's completely weird. Right. And it has to be vital to us, otherwise we'll have evolved not to need it. Right, and you're so vulnerable. Like a sleep. Yeah, you're not beating, procreating, and you're in danger. Right. So I think a lot about what this might be like. And as a result, I carry some scientific tools to present to an alien in the first encounter. Because that would be language we have in common. Science and math, yeah. But otherwise, yeah, Pythagoras. Pythagoras theorem, most certainly, yeah. So, yeah, I'm sad that I've not met aliens, but it's not for want of trying. There's a book called... Where to find them? No, a book, How to Avoid Being Abducted by Aliens. Just be. How to avoid an alien abduction. I read that book cover to cover. And I disobeyed every...

Destruction within it. And they still didn't want to. Don't put it that way, Chuck. So, and that was of course Emily Blunt, who was in Devil Wears Prada as Emily. And she was also Mary Poppins, one of my favorite characters. Oh, OK. Big shoes to fill. I know, I know. Big shoes to fill. No, she's great. And dare I say. Damn, damn. Dare I say, and I'm not one to do this, I think she gave an Academy Award-winning performance in Disclosure Day. Wow. Given the range of acting she needed to display.

Wow. She's inhabited by the spirit of these aliens and so she, her character is different. I got to see the movie. Yeah, yeah, yeah. Very cool. Emily Bunt, thank you. Emily. Guys, we got time for one more. All right, better, better, better. Here we go.

Hi, I'm Kevin Sussman. I play Stuart on Stuart fails to save the universe. Here's my question for you, Neil. If time travel were possible, could it be used to travel long distances using quantum physics? Okay, so that's that is the Stuart character that was just lifted up from Big Bang Theory and this next show pivots on him. And his exploits and he brings his friends and well, I'm not according to that question. It doesn't just sounds like he wants us So if you can time travel right then It doesn't matter how long it takes you to get somewhere. No You just time travel back into your past and you're back to where you started where you started but in a different place right seems to me It's kind of cool be kind of cool using kind of like a Stargate portal You could, but I'm just saying, even if you didn't, you just go back in time to when... Right. You didn't burn all those years. Yeah. That'd be great. Yeah, yeah. So what's the theory about traveling back and traveling forward in time? Yeah, so we know how to travel forward in time. We don't really know how to travel backwards in time. Although Richard Gott III wrote a book called Time Travel in Einstein's Universe.

It's in our archives. You can dig it up. The conversation I had with him. He's a friend and colleague. We've co-written three books together back when I was at Princeton and germinated back from back when I was in Princeton. So traveling backwards, you get the paradox problem. And hence, Stephen Hawking suggested one day we will stumble on or find on purpose the time travel protection conjecture. which prevents you from traveling back in time to kill your parents, which would then prevent you from having been born to go back in time to kill your parents. That's a paradox that you want to avoid. So either it's not possible at all, or there's some way to do it but using other universes. No, that makes sense. That's kind of cool.

Yeah, if only James Cameron... I'm glad nobody told James Cameron this, or we would have never had the Terminator. Plus, you don't have to kill the people. You just have to prevent them from having sex in the moment that would have produced you. True. Have sex ten minutes later. Right. And

guess what? And you'd probably be another you. Yeah. Wow. So, he's the geekiest one of the whole series. Yeah. Well, I might have to check out still where it fails to save the universe. The only show about failure that is good to watch. So, I think this whole time we have. We have more though. We have more. We have plenty. I'm sure we've got plenty of more stars than when I ask questions. All right, so. I'm fine. Always good to have you guys. Likewise. This is Neil deGrasse Tyson. Start Talk Special Edition right here. From my office at the Hayden Planetarium. Until next time. Keep looking up.